

Pharmacology of Antiarrhythmic Drugs

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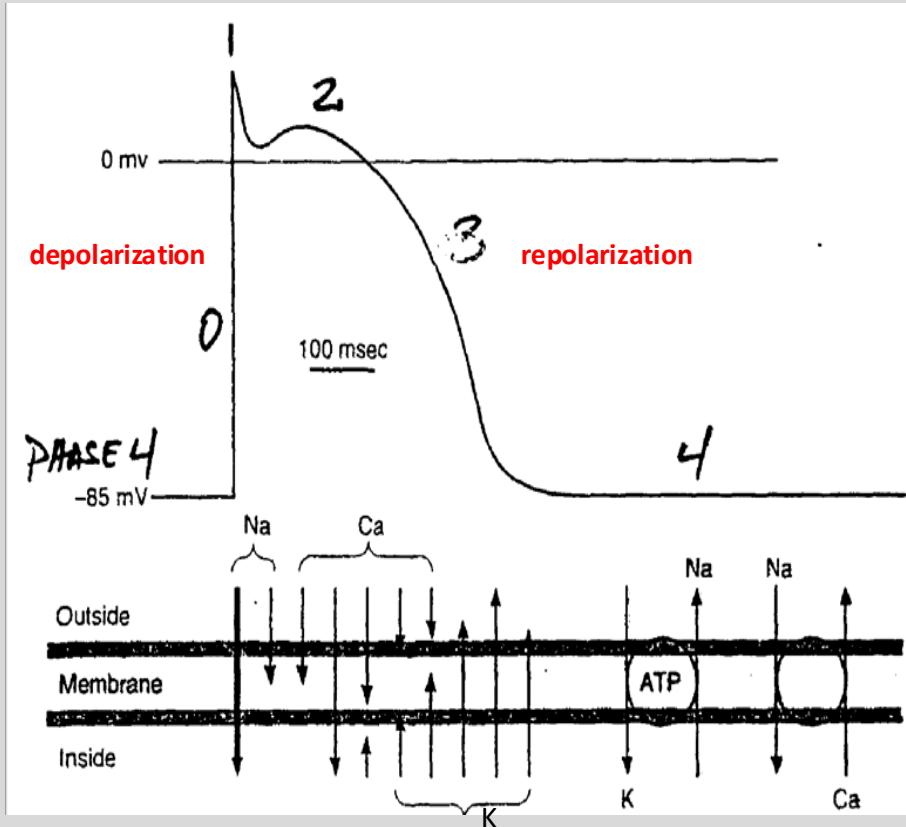
Session Objectives

- **Recognize and categorize antiarrhythmic drugs**
- **Describe the electrophysiological mechanisms of the antiarrhythmic drug classes**
- **Describe the specific adverse effects associated with these drug classes**
- **Describe the uses of each group of antiarrhythmic. Include contraindications.**
- **Describe the miscellaneous drugs used to treat arrhythmias.**

Cardiac Action Potential

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Phase 0: Opening of sodium channels and entry of sodium into the cell via ion channels and down a concentration gradient. “upstroke”

Phase 1: Sodium channels close. “initial repolarization”.

Phase 2: Calcium channels open and allow calcium entry “plateau”

Phase 3: Potassium channels open as calcium channels are closing “repolarization”

Phase 4: Potassium remains the major conductive ion as the sodium-potassium pump resets the concentration gradient for each ion.

Na⁺-Ca²⁺ antiporter re-establishes the appropriate Ca²⁺ concentrations.

EKG

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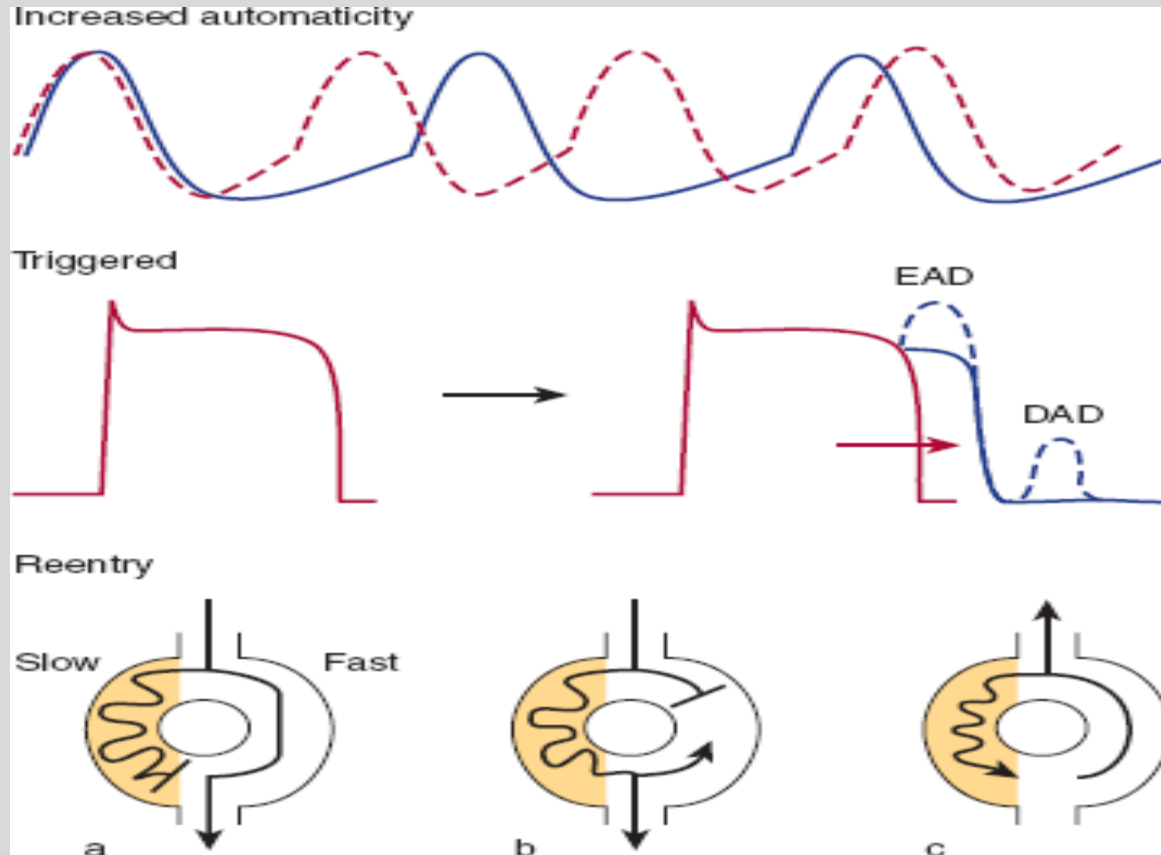


- **P wave: atrial depolarization**
- **PR: AP thru atria and AV node**
- **QRS: ventricular depolarization**
- **T wave: ventricular repolarization (K⁺ levels)**
- **QT: duration of ventricular AP**

Mechanisms of Arrhythmias

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Kusumoto FM. Cardiovascular Disorders: Heart Disease. In: Hammer GD, McPhee SJ. eds. Pathophysiology of Disease: An Introduction to Clinical Medicine, Seventh Edition New York, NY: McGraw-Hill; 2013.

Classification of antiarrhythmic drugs (Vaughan-Williams)

- **Class 1: Sodium channel blockade**
- **Class 2: Sympatholytic; β -adrenergic-blockade in the heart**
- **Class 3: Potassium channel blockade**
- **Class 4: Calcium channel blockade**
- **Class 5: Miscellaneous (digoxin, adenosine)**

Class I Antiarrhythmic Drug Effects

On the Ventricular Action Potential:



On the ECG:

↑QRS & ↑QT

↓QT

↑↑QRS

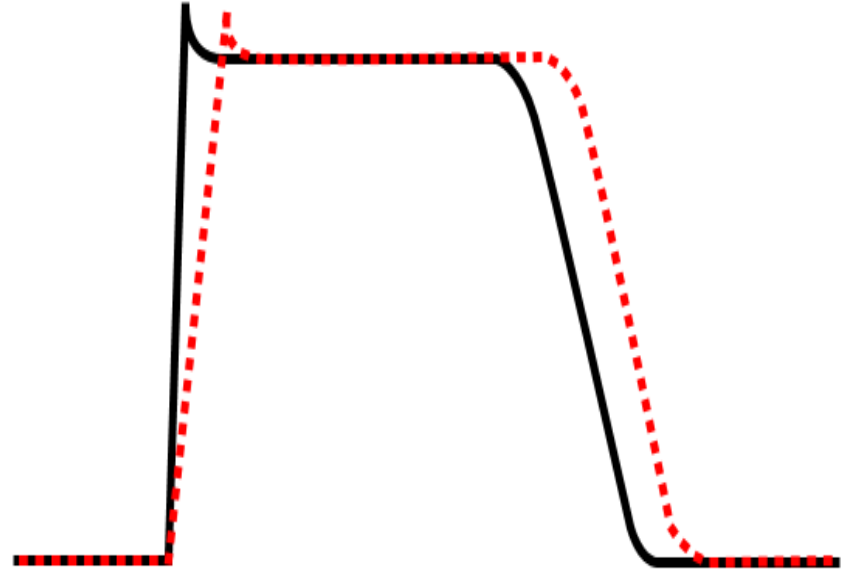
Class I: Use Dependent

- **Drugs inhibit depolarization in tissues that are more frequently depolarizing.**
- **Increased sodium channel blockade at faster rates (tachycardia).**

Class Ia Drugs :

The Queen, Prefers, Discos

- Slow the rate of rise of phase 0 and prolongs the ERP of the ventricle.
- Works on open channel
- Some K⁺ channel blockade
- Treats atrial and ventricular arrhythmias
- Prolongs QT and QRS
- Quinidine, Procainamide, Disopyramide



Quinidine

- QT-interval prolongation; torsades de pointes
- M₂ blockade
- Cinchonism: headache, dizziness, and tinnitus. Like quinine (treats *P. falciparum*)
- Immunologic reactions: thrombocytopenia, hepatitis.

Procainamide

- QT-interval prolongation; torsades de pointes
- M_2 and alpha blockade
- Acetylation (N-acetylprocainamide, NAPA)

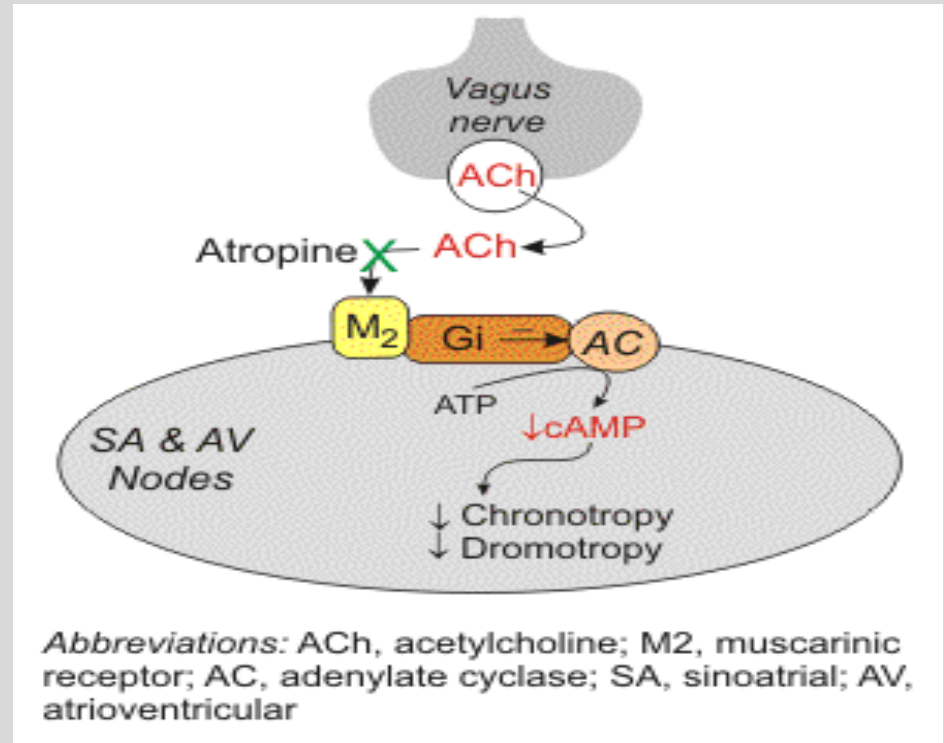


Disopyramide

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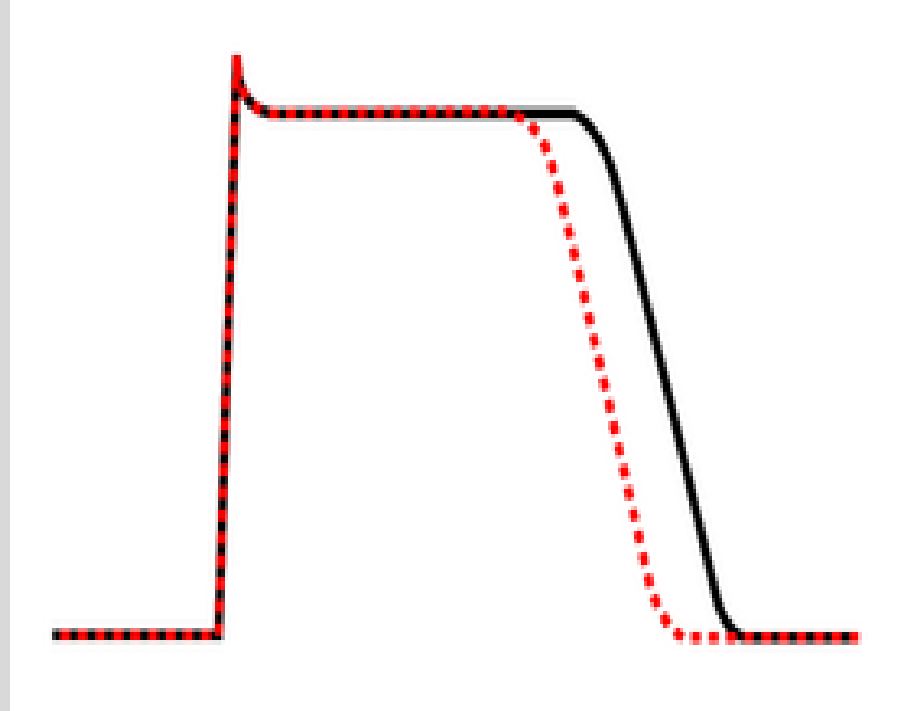
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- Most “atropine-like” activity”
- Adverse effects??
- Administer with an AV node suppressant



Class Ib Drugs: Lidocaine, Mexiletine

- Minimal effect on phase 0, shorten the AP and ERP
- Works on activated and **inactivated** channels
- Rapid dissociation from channel
- Prevents ventricular arrhythmias post MI
- Decreased QT interval

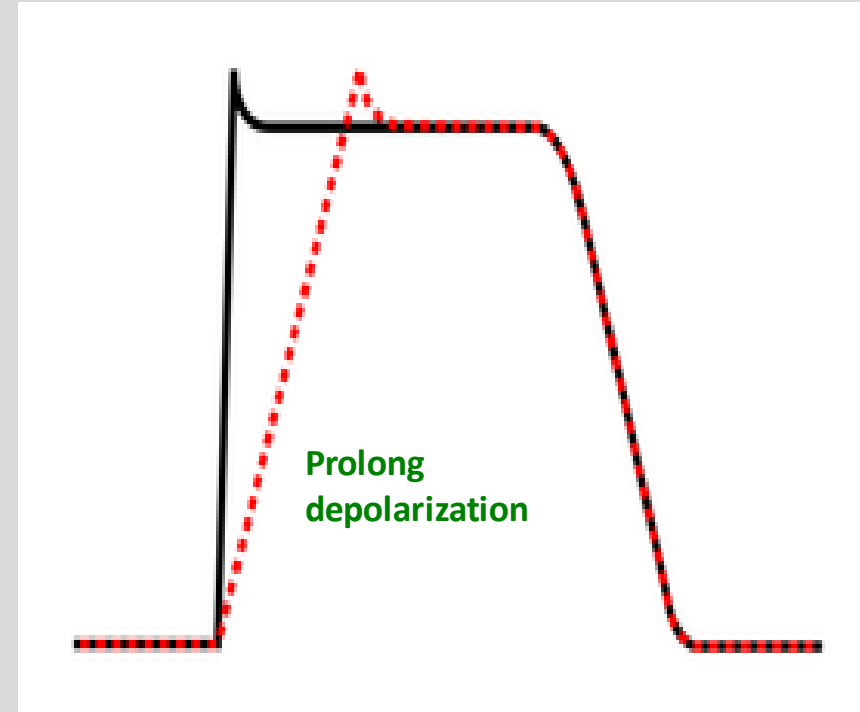


Lidocaine, Mexiletine

- Lidocaine (IV)
- Inactivation in **depolarized** cells reduces conduction.
- Local anesthetic; produces **neurological** adverse effects. Substrate for CYP450.
- Mexiletine (po) and phenytoin; Similar MOA
 - Mexiletine: GI adverse effects
 - Phenytoin: hirsutism, gingival hyperplasia, nystagmus

Class Ic Drugs: Flecainide, Propafenone

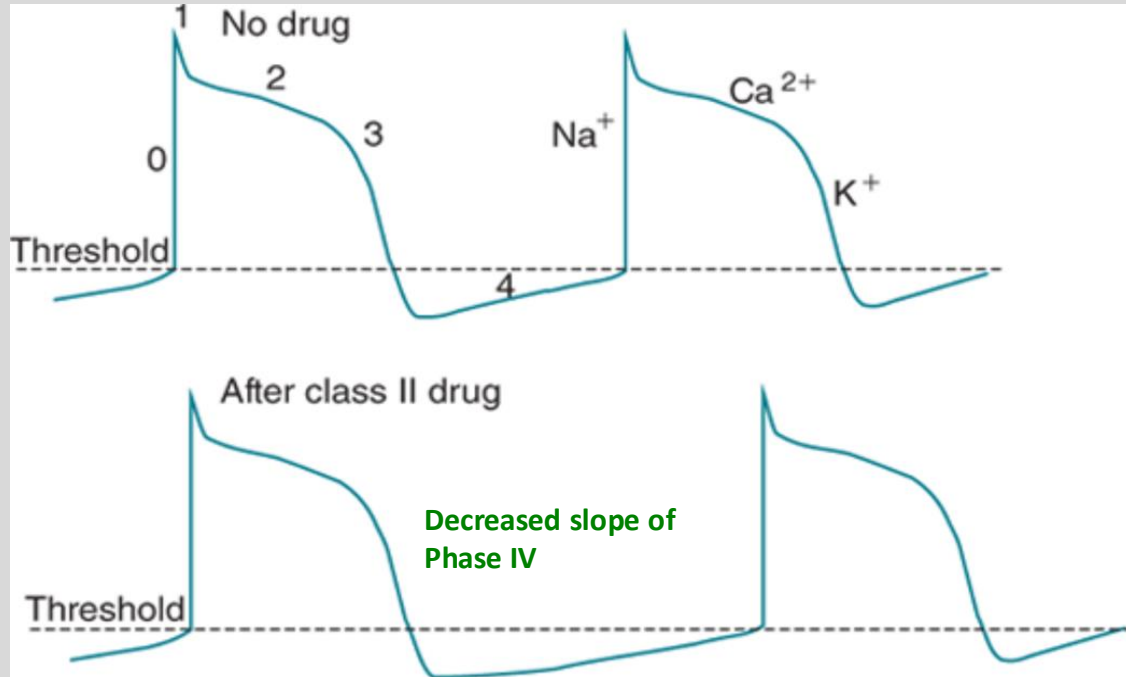
- Minimal effects on the APD
- Dissociate from the channel **slowly**
- Works on **open** channel
- Treats refractory ventricular arrhythmias in patients without structural damage. Also, for atrial fibrillation. CAST***
- Increase QRS. Can increase ventricular rate.
- Negative inotropic effects (worsens CHF)
- Flecainide: blocks ryanodine receptor (CPVT treatment)



Class II: Beta-receptor antagonists

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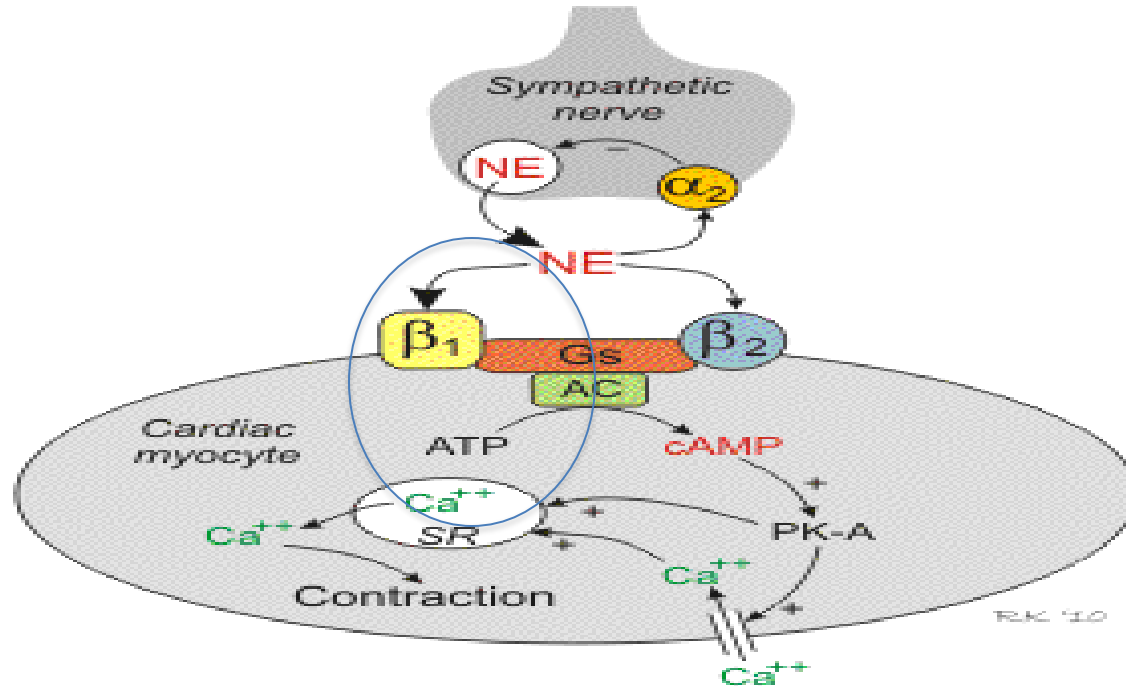


Source: Janet L. Stringer: Basic Concepts in Pharmacology: What You Need to Know for Each Drug Class, Fifth Edition, www.accesspharmacy.com
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Beta-receptor antagonists

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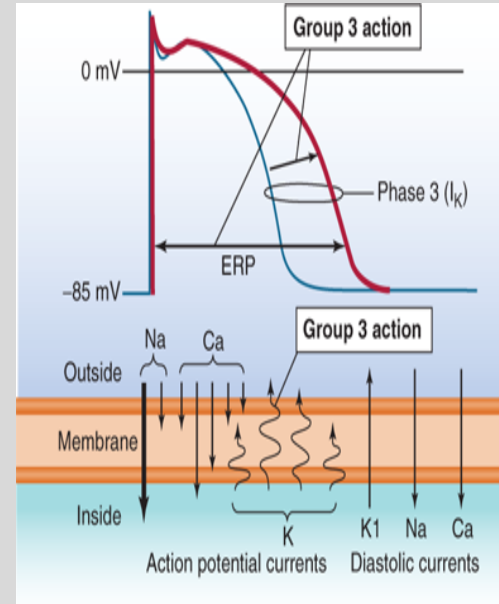
Abbreviations: NE, norepinephrine; Gs, G-stimulatory protein; AC, adenylyl cyclase; PK-A, cAMP-dependent protein kinase; SR, sarcoplasmic reticulum

Beta-receptor antagonists

- Beta 1: decrease HR, CO, and renin from JXA. **Suppress AV and SA nodes. Increase refractory period and AP duration**
- Use: hypertension, MI, CHF, arrhythmia, thyrotoxicosis
- Beta 1: (atenolol, metoprolol)
- Beta 1, 2: (nadolol, sotalol, propranolol, timolol). Bronchoconstriction, hypoglycemia, PVD (Beta 2 block)
- Pindolol, acebutolol (ISA). **Worsens condition**
- Esmolol (ultra-short acting)

Class III: Potassium Channel Blockers

- Delay repolarization
- Prolong AP and ERP without altering phase 0 depolarization or the RMP.
- Treats: A.fib, atrial flutter, ventricular tachycardia
- Prolongs QT (torsades)
- “Reverse-use dependence”
- Amiodarone, ibutilide, dofetilide, sotalol



Source: A.J. Trevor, B.G. Katzung, M. Kruldering-Hall: Katzung & Trevor's Pharmacology: Examination & Board Review, 11th Ed.
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Amiodarone

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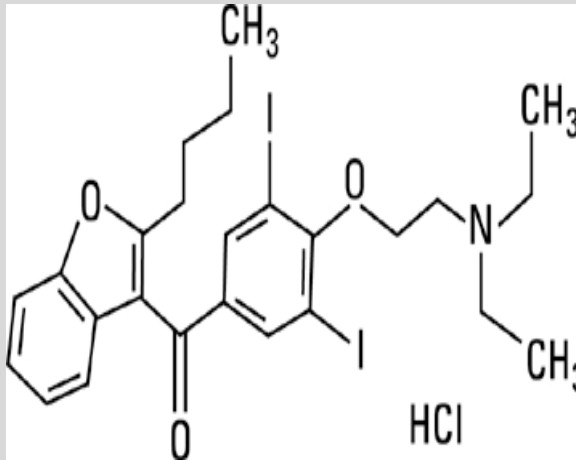
- **Broad-spectrum; blocks potassium, calcium, and sodium channels. Some beta and alpha-adrenergic blockade.**
- **Lower incidence of increased QT vs Class Ia drugs**
- **Substrate of CYP3A4 (drug interactions)**
- **Also inhibits P450 enzymes (digoxin, warfarin, statins)**
- **Long half life (~60 days)**
- **Desethylamiodarone (hepatotoxic)**

Amiodarone

- Stops conversion of T_4 to T_3 (hypothyroidism)
- Iodinated (hyperthyroidism)
- “Smurf-skin”
- Corneal deposits

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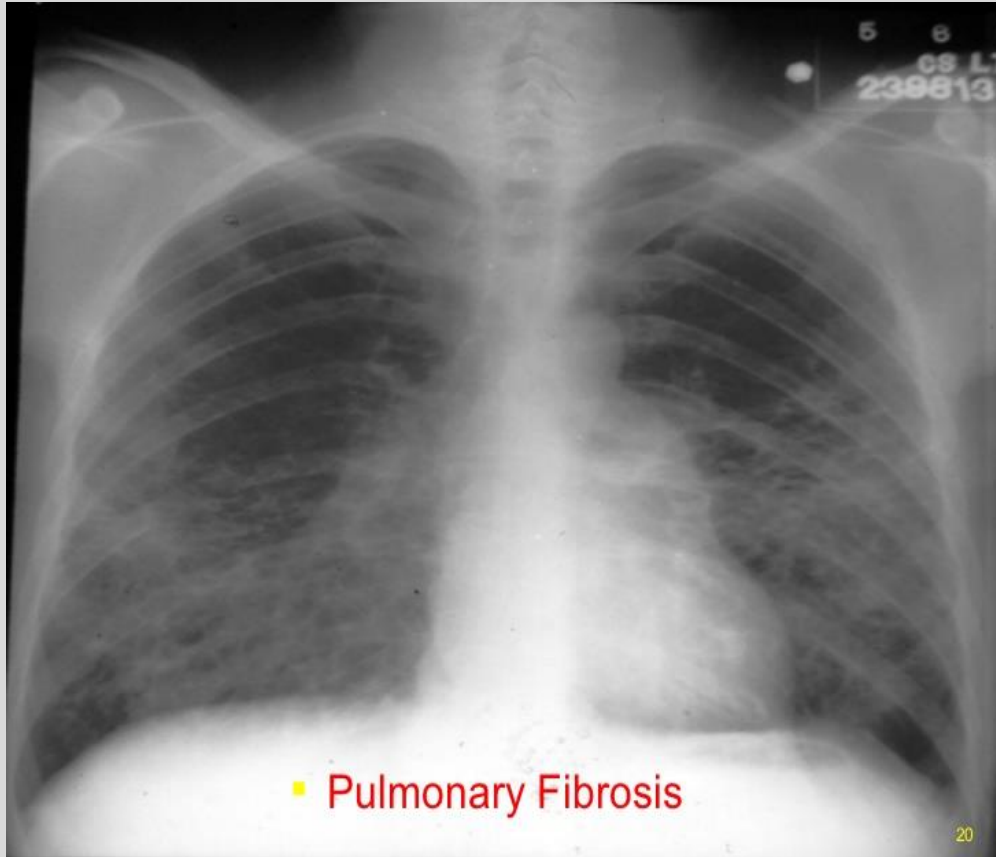
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Amiodarone

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- **Baseline and yearly chest X-rays are recommended for patients starting amiodarone, and in some cases, pulmonary function tests.**
- **Management: D/C drug; Administer glucocorticoids**

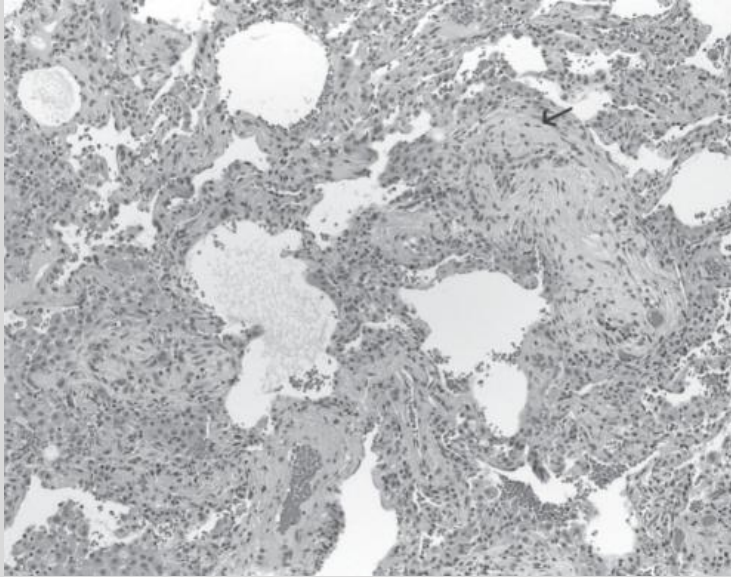
Review: Other drugs that cause pulmonary fibrosis

- Bleomycin
- Busulfan
- Methotrexate
- Nitrofurantoin

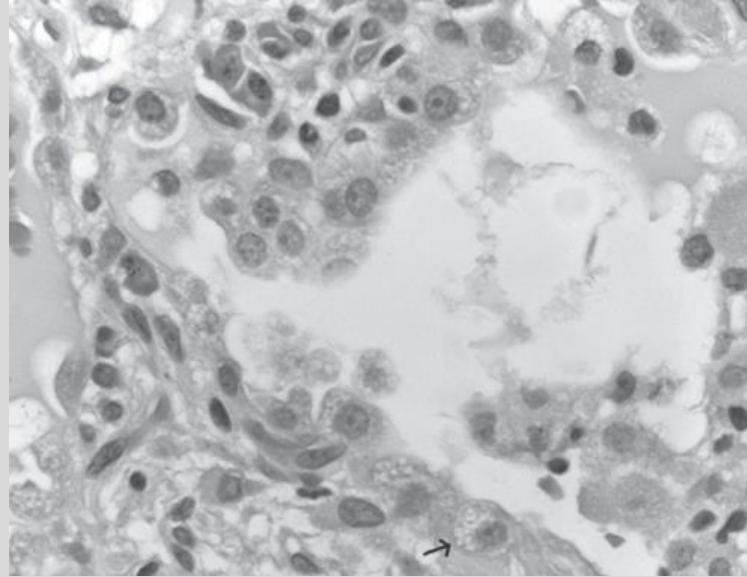
Amiodarone and Lung Tissue

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Interstitial inflammation



Hyperplasia of Type II pneumocytes

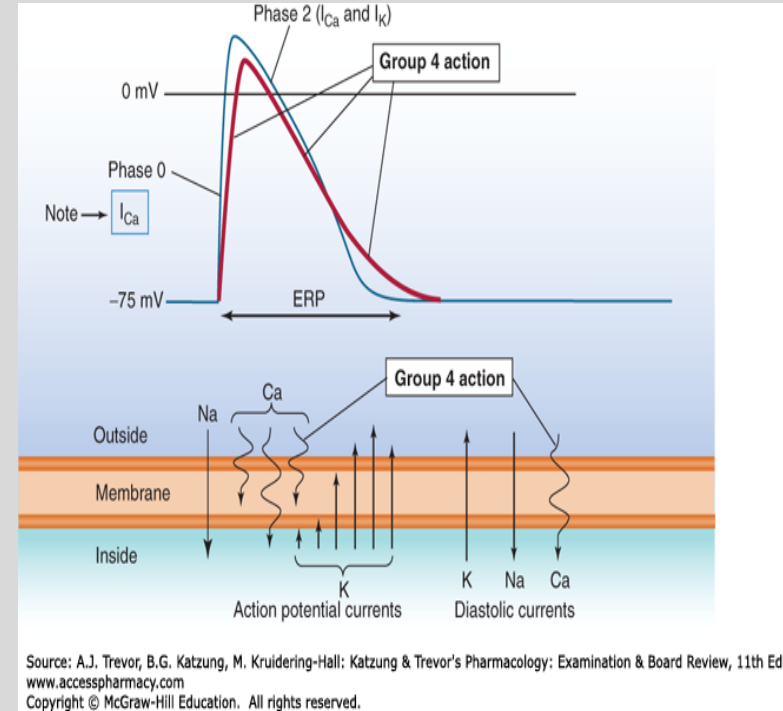
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2687560/>

Class III Drugs

- **Dronedarone:** (like amiodarone, with no I⁻)
- **Sotalol:** non-selective β -adrenergic receptor-blocker (II) and increases AP (III). Also, for atrial fibrillation treatment.
- **Dofetilide, ibutilide:** slow cardiac repolarization. Used for management of atrial fibrillation and flutter. Must monitor EKG for 4 hours.

Class IV: Calcium Channel Blockers

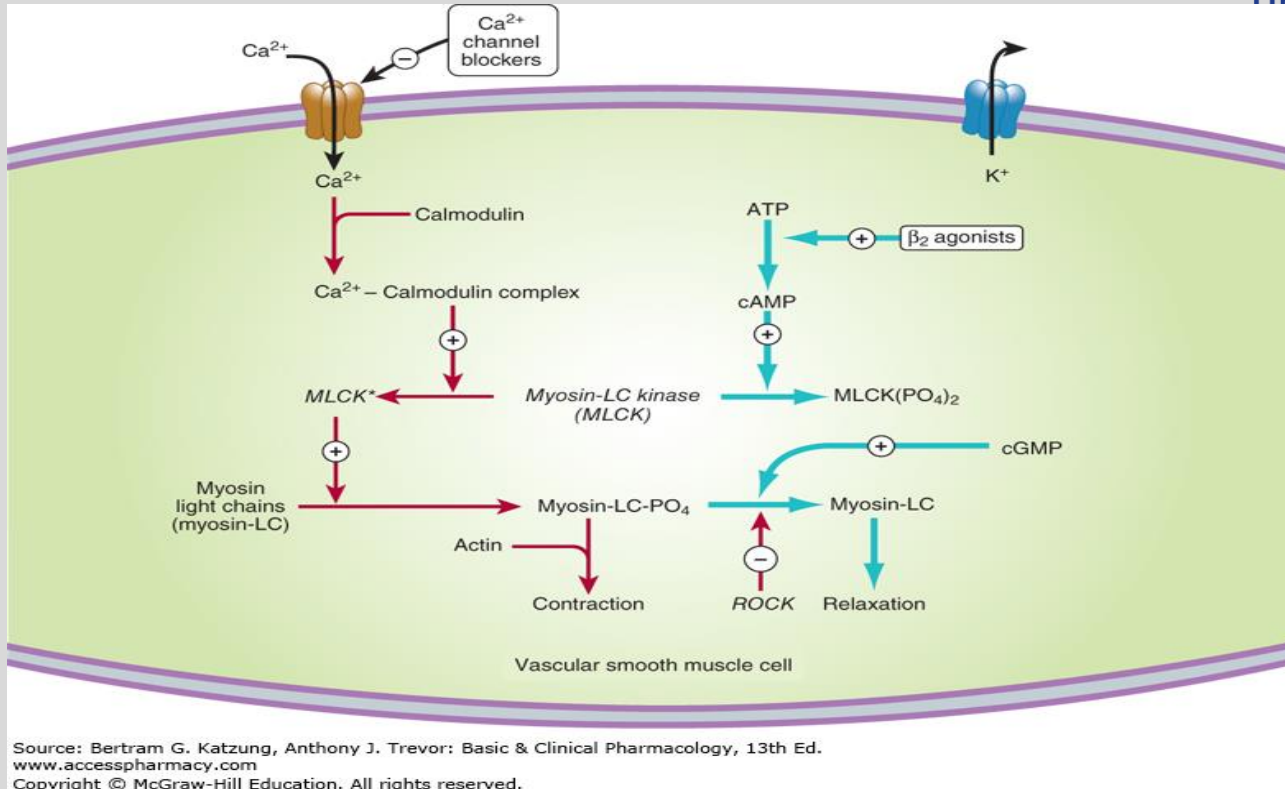
- Increase ERP, especially in AV node
- Slow inward calcium current between phase 0 and 2
- Decrease phase IV in SA, AV node (like Beta blockers)
- Prevent SVT, atrial fibrillation
- **Verapamil** and **Diltiazem**.
NOT -dipines



Calcium Channel Blockers

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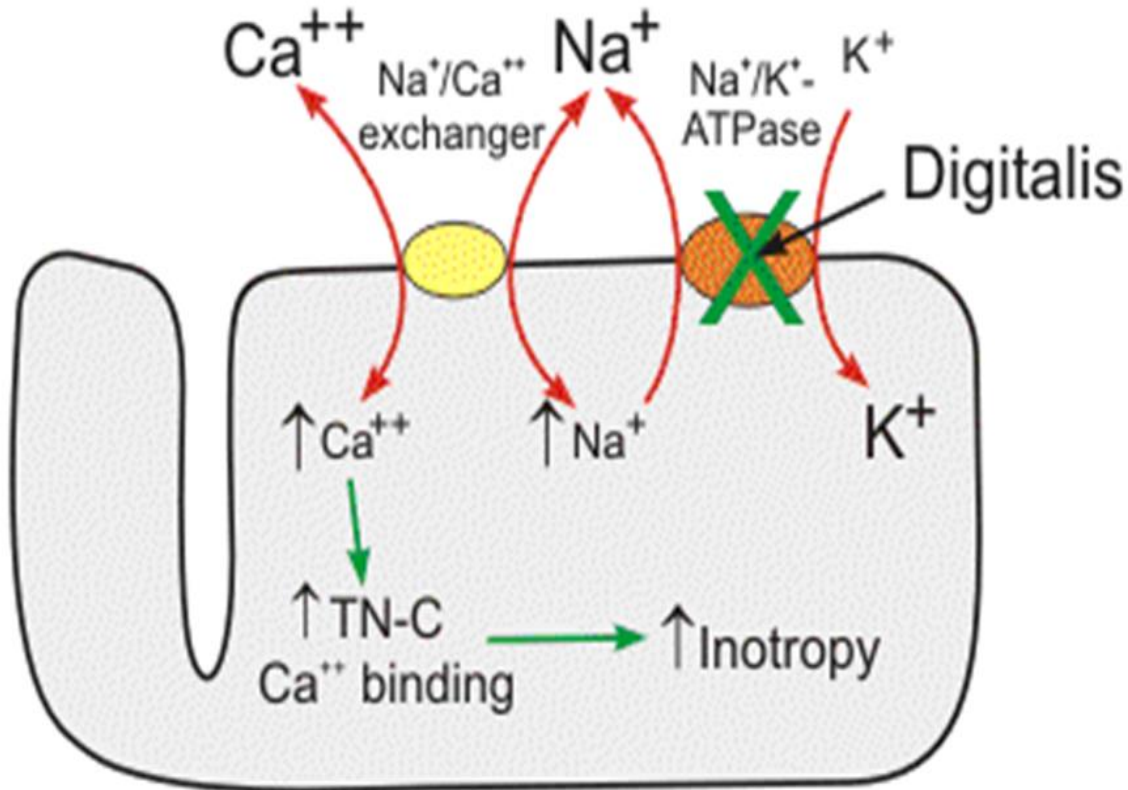
Calcium Channel Blockers

- Arteriole and Heart; Verapamil > Diltiazem
- Resist the SANS reflex, can be used in place of beta blockers
- Use: Hypertension, angina, arrhythmias (SVT, Class IV)
- Adverse effects: Dizziness, headache, edema, tachycardia (dihydropyridines). Bradycardia, constipation (most with verapamil), gingival overgrowth, grapefruit drug interaction.

Digoxin

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Digoxin (Digitalis)

- Long $t_{1/2}$
- Negative chronotropic:
increases parasympathetic tone
(vagal)
- Positive inotropic: increases
contractility
- Increased afterdepolarizations
(monitor patient for low
potassium)
- Inhibition of (Na⁺/K⁺ ATPase)



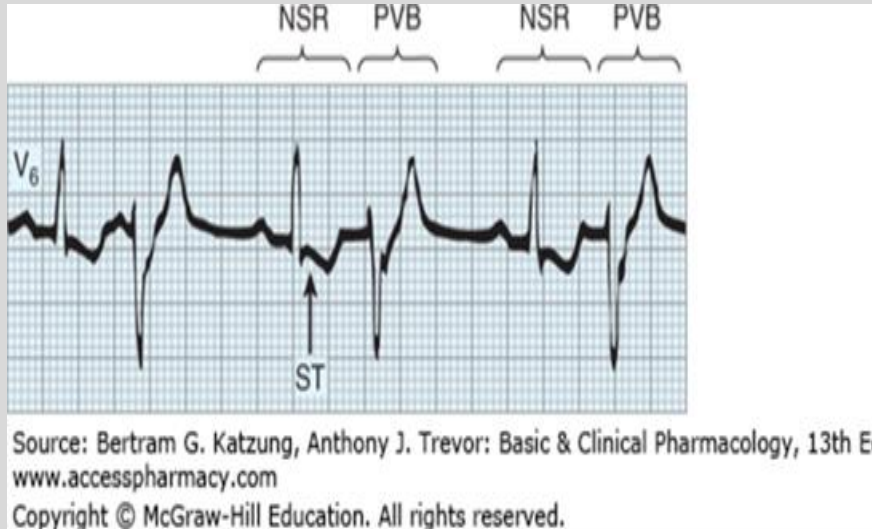
Digoxin

- N/V/D, anorexia
- Ventricular arrhythmias: High Ca, low K, Mg (monitor diuretic use)
- Yellow vision (halos)--Xanthopsia
- Gynecomastia (estrogenic)
- DI: antacids, cholestyramine (absorption) amiodarone (CYP inhibition), verapamil and quinidine (elimination)
- Renal impairment (reduced digoxin elimination)
- Antidote: DigFab, manage electrolytes. Give magnesium (torsades)



Digoxin

- Increases PR interval (reduced AV node)
- ST depression (hockey stick configuration)



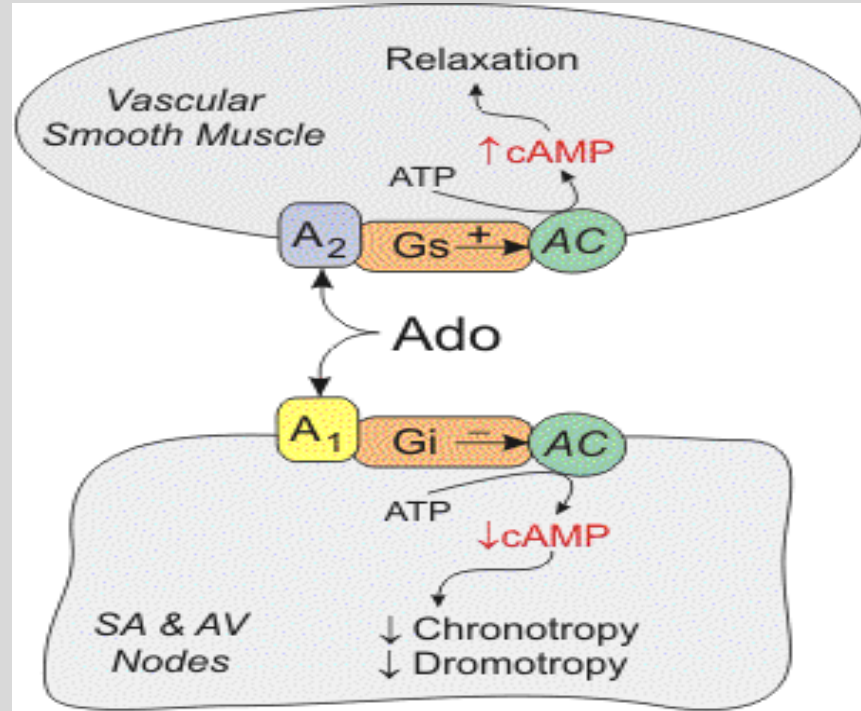
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Adenosine

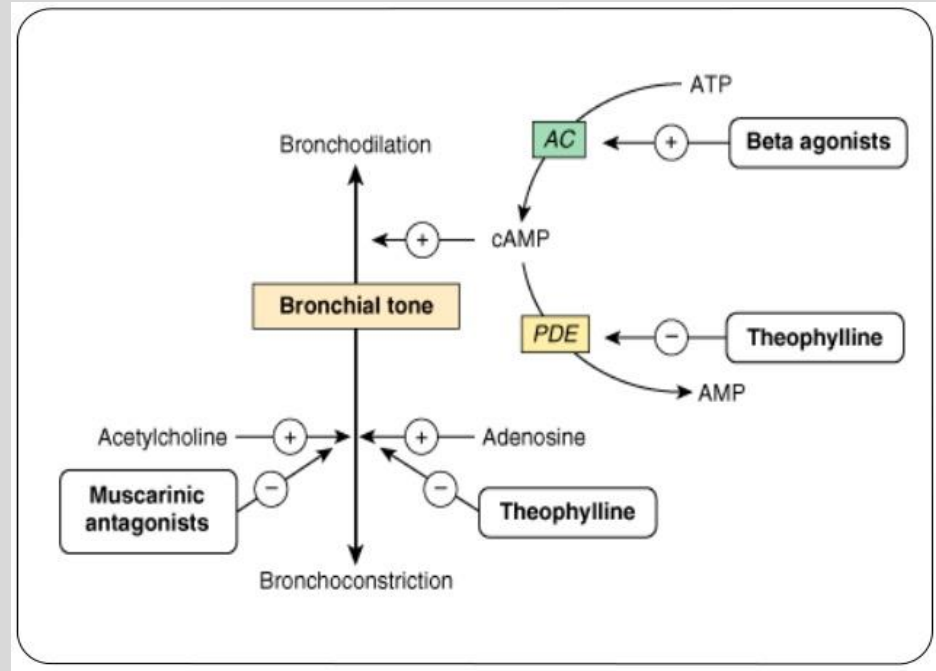
- Short half life (~5 sec)
- PSVT treatment
- Gi (A₁ receptor), decreased SA and AV node action
- Gs (A₂ receptor), vasodilation
- Gq (A_{2B} receptor), bronchoconstriction

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Adenosine

- Hypotension
- Bronchoconstriction
- “Opposite” to theophylline in mechanism (Methylxanthine)



AV nodal suppressants

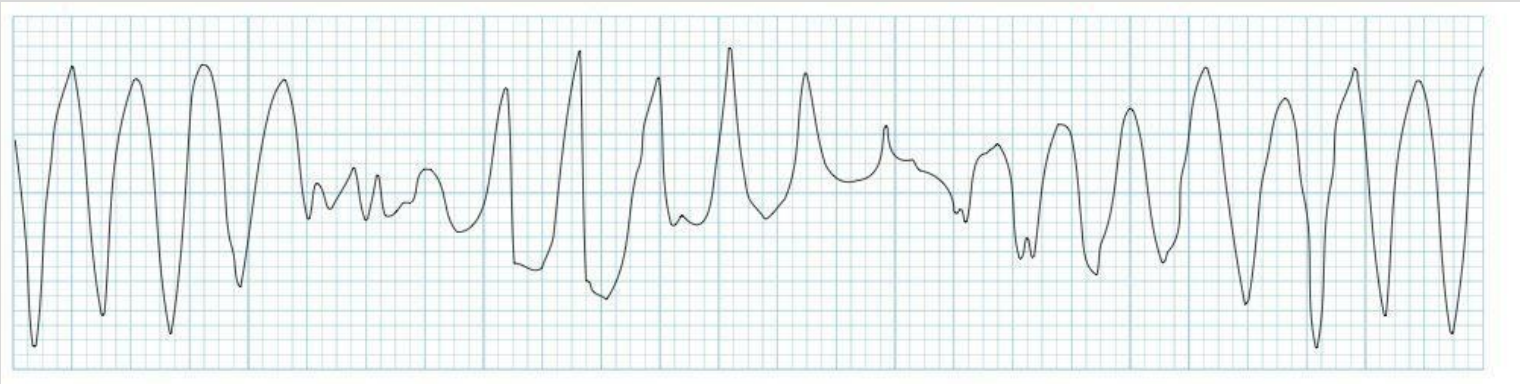
- **Calcium channel blockers (Class IV). Verapamil and Diltiazem**
- **Adenosine (Gi mechanism, for PSVT)**
- **Digoxin (vagal, parasympathetic)**
- **Beta-adrenergic receptor antagonists (Class II)**

Magnesium

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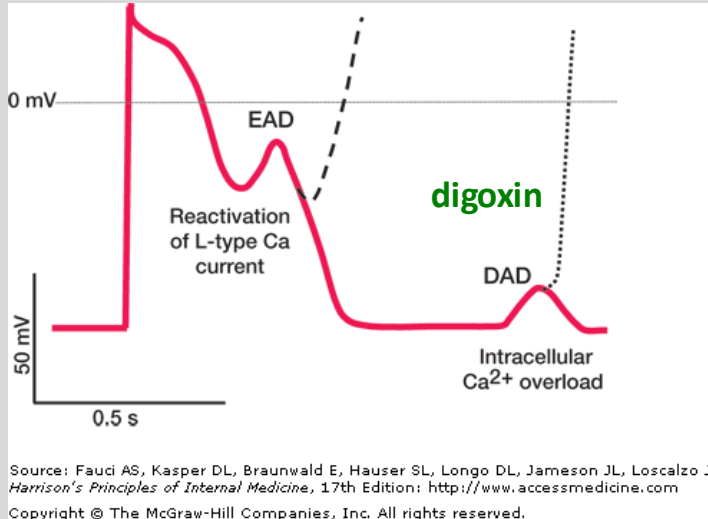
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- For digitalis-induced arrhythmias and torsades.
- Inhibits calcium flux into cytosol.



Potassium

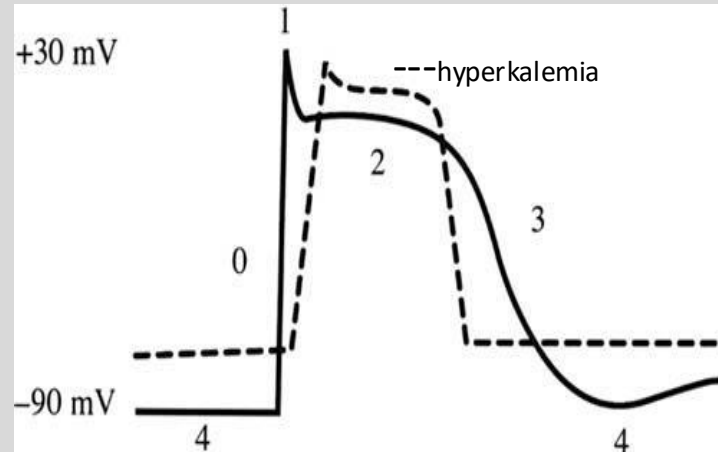
- Hypokalemia: Slowed repolarization.
- Early and late after-depolarizations.



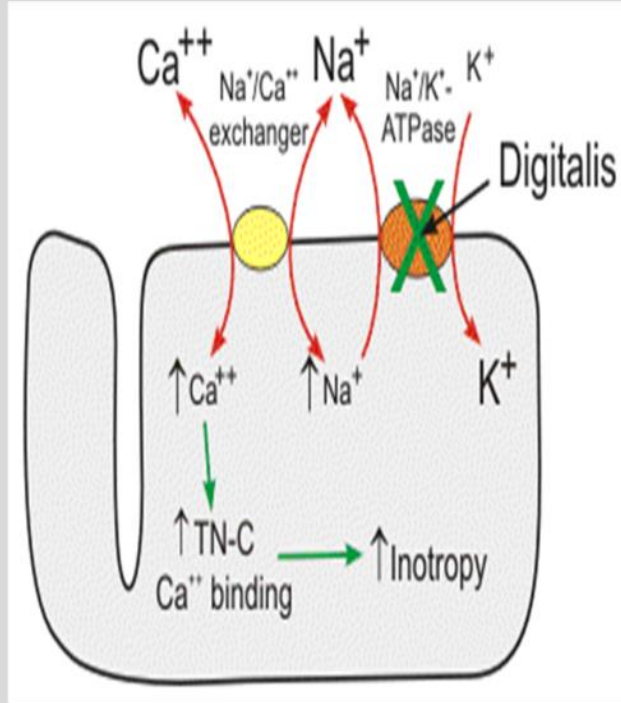
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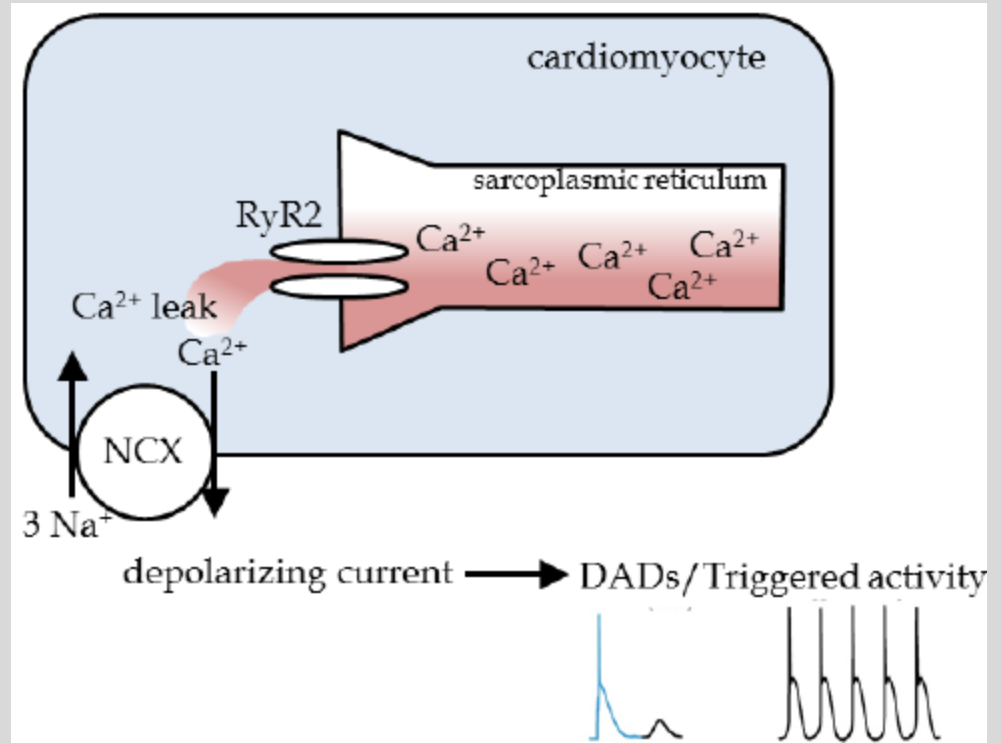
Hyperkalemia: Depressed pacemaker and slowed conduction. Decreased AP duration.



Digoxin



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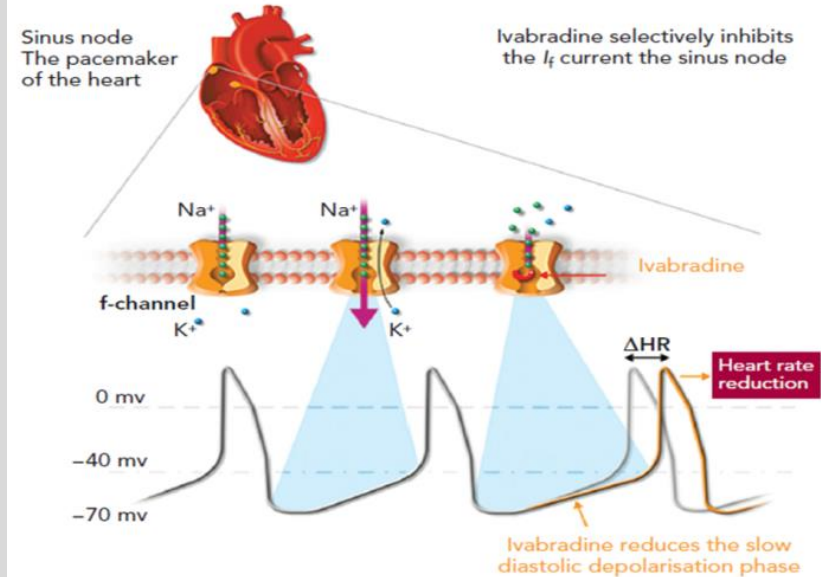


Ivabradine

- Selective I_f inhibitor (“funny channel” blocker)
- Inhibits action in the SA node without affecting the duration of the action potential
- Use: CHF and stable angina, especially those who can’t tolerate or be managed with beta antagonist
- Adverse effects: bradycardia, vision changes, not in pregnancy. Caution with CYP 3A4 inhibitors

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Figure 1: Mechanism of Action of Ivabradine

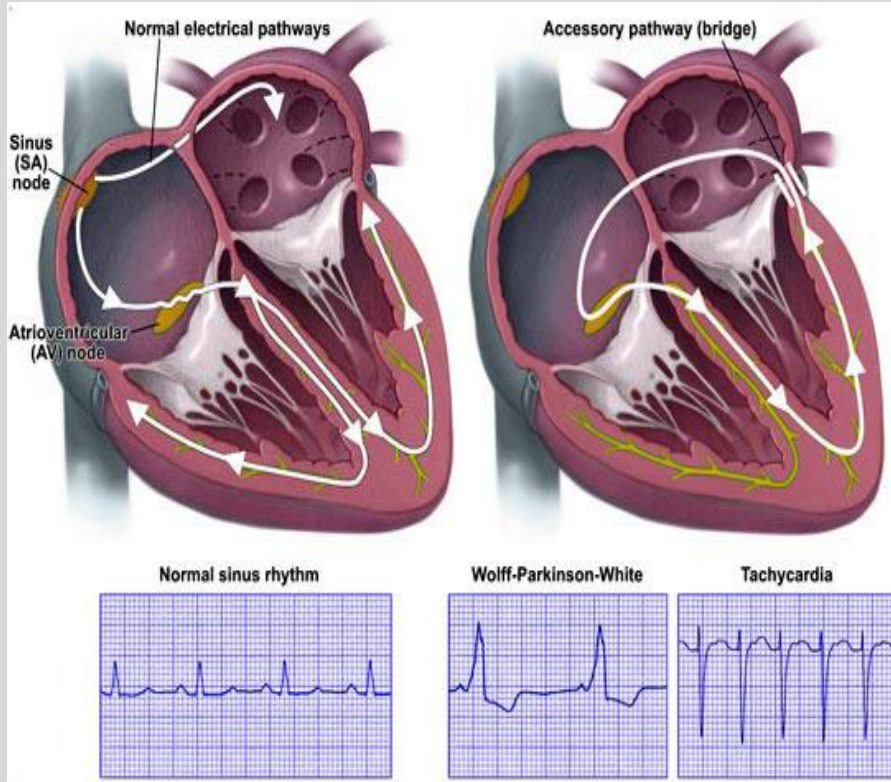


Source: <http://www.shift-study.com/ivabradine/mode-of-action/> Reproduced with the permission of Servier © 2016.

Wolff-Parkinson-White

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- AV node suppressant not used for WPW in patients with atrial fibrillation or antidromic AV reentry tachycardia
- Use Procainamide (inhibits initiation) or Amiodarone (increases refractoriness)

Delta wave

Drug Indications

- **Sinus tachycardia: Class II, IV**
- **Atrial flutter: Class Ia, Ic, II, III, IV, or digoxin**
- **PSVT: adenosine**
- **AV block: atropine (M_2 blockade)**
- **Ventricular tachycardia: Class I, III, use Ib if ischemia**
- **Ventricular fibrillation: Class Ia, Ib, III**
- **Torsades: Magnesium, Class II, IV.**
- **Tricyclic-antidepressant overdose: Sodium bicarbonate**

Summary Slide

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- **Correlate the electrophysiology of the cardiac action potential with antiarrhythmic drug mechanisms**
- **Describe the MOA and adverse effects of these drugs.**

Core References:

- **Katzung & Trevor Basic and Clinical Pharmacology, 16e, 2024; Chapter 14, Agents Used in Cardiac Arrhythmias**
- **Pharmacology of Antiarrhythmic Drugs Summary (Maria A. Pino, Ph.D)**
- **Scholar Bricks: See CPG**

Review Question

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A 65-year-old female presented to the clinic with chest pain on exertion. She has a history of type II diabetes and smokes two packs of cigarettes daily. Her pulse is 74/min and blood pressure is 155/105 mm Hg. Serum studies revealed an LDL cholesterol of 180 mg/dL. She was prescribed a medication that inhibits the hepatic production of circulating molecules implicated in the pathogenesis of coronary artery disease. The concomitant use of gemfibrozil with this medication would increase the incidence of:

- A. Bradycardia
- B. GI bleed
- C. Hyperkalemia
- D. Metabolic alkalosis
- E. Myoglobinuria
- F. Pancreatitis

Lecture Feedback Form:

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<https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R>